Introduction to Software Engineering

Software Design



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Software Design

Objectives

This document focus on the following topics:

- ✓ Completing the Software Design document with the following sections:
  - Architectural design (C4, ADR, deployment)
  - Object-oriented design (domain model, class and sequence diagrams)
  - Data design
  - User interface design
  - Traceability update
- ✓ Including an AI usage log for this phase
- ✓ Understanding the Software Design document.

1 Member Contribution Assessment

[List every member. Contribution (%) must add up to 100%. Evidence: links to tasks on the board, commits or pull requests.]

ID	Name	Role(s)	Contribution (%)	Evidence
<Student1>				
<Student2>				
<Student3>				
<Student4>				
<Student5>				

2 Design Overview

2.1 Design Goals and Constraints

[List the architecturally significant requirements (key NFRs and high-priority functions) that drive the design, and any constraints (technology, budget, deadline).]

3 Architectural Design

3.1 Architectural Style

[Name the architectural style(s) used (layered, MVC, client--server, microservices, event-driven, ...) and justify the choice against the design goals.]

3.2 System Context Diagram (C4 level 1)

[Update the context diagram from PA1.]

3.3 Container Diagram (C4 level 2)

[Show web/mobile apps, API/back end, database, external services and the technology of each container.]

3.4 Component Diagram

[Decompose the main container(s) into components/modules and show their dependencies (system decomposition).]

3.5 Deployment Diagram

[Show where each component runs (devices, servers/cloud services, database server) and how they communicate.]

3.6 Architecture Decision Records

[Record at least 3 important decisions (e.g., database, framework, authentication, hosting). One table per ADR; store them in `/docs/design/adr`.]

ADR	ADR-001: <title>
Status	proposed | accepted | superseded
Context	
Decision	
Alternatives considered	
Consequences	

4 Object-Oriented Design

4.1 Domain Model

[Conceptual class diagram of the main business entities and their relationships (no methods).]

4.2 Class Diagram

[Design-level class diagram: class names, relationships (type, multiplicity). Include attributes and operations if the diagram is not too large; split by package if needed.]

4.3 Class Specifications

[Specify the 6--8 most important classes.]

4.3.1 Class C1

[Superclass (if any), responsibility of the class.]

#	Attribute	Type	Visibility	Constraint	Description

#	Operation	Parameters / return	Visibility	Description

4.4 Sequence Diagrams

[Draw a sequence diagram for the main flow of 3--5 key use cases (include at least one alt fragment). Every message should correspond to an operation in the class diagram.]

4.5 State Machine Diagrams

[For objects whose behaviour depends on their status (order, booking, account, ...), draw a state machine diagram.]

4.6 Design Principles and Patterns

Principle / pattern	Where it is applied	Why
<e.g., Strategy>	<ticket pricing>	<different discount rules>

5 Data Design

5.1 Data Diagram

[ERD / database diagram of the stored data and their relationships (MySQL Workbench, dbdiagram.io, ...).]

5.2 Data Specification

[For a DBMS: describe each table. For JSON/XML/file storage: describe the structure, data types, constraints and give a sample file.]

Table	Column	Data type	Constraints (PK, FK, NOT NULL, UNIQUE, CHECK)	Description

5.3 Normalisation and Indexes

[State the normal form reached (at least 3NF, or justify denormalisation) and the indexes created for frequent queries.]

6 User Interface and User Experience Design

6.1 Screen List and Screen Flow

[Draw the screen flow diagram (navigation between screens).]

#	Screen	Description	Related use case

6.2 Screen Specifications

[Specify the 5--6 most important screens: design image, controls, events and validation. For the other screens, the design image is enough.]

6.2.1 Screen ``A''

#	Control	Type	Description / validation	Event handling

6.3 Accessibility Checklist

- Text contrast $\geq 4.5{:}1$; colour is not the only way to convey information
- All functions usable with the keyboard; visible focus
- Form fields have labels; error messages explain how to fix the problem
- Layout works on mobile and desktop (responsive)

7 Traceability Update

[Extend the traceability matrix from PA2 with the design elements (component, class, screen) that realise each requirement.]

8 Changes to the Requirements Since PA2

[List any changes made to the PA2 requirements after submission: what changed, why, and which items were affected.]

#	Change	Reason	Affected items

9 AI Usage Log

[Required in every PA (course GenAI policy). List each significant use of AI tools. Write ``No AI tools were used'' if none. Undisclosed AI use is treated as an academic integrity violation.]

#	Member	Tool	Task / purpose	How the output was verified and changed
1		<e.g., ChatGPT, Claude, Copilot>	<e.g., draft use case UC-03 specification>	<e.g., reviewed with the BA, corrected 2 alternative flows>